

Software Engineering

PRISM

Formal Modeling and Analysis

- Exercises -

Exercise 1

- Source: Dynamic Power Management (DPM) case study available from <http://www.prismmodelchecker.org/courses/pmc1112/>
<http://www.prismmodelchecker.org/tutorial/>
 - Start from (P, S and R) properties given in the course lecture slides
1. Add a reward structure to the model called "lost", which assigns 1 to every transition of the model labelled with action "request" from a state where the queue is full. Next, create a new property to check the expected cumulated reward up until time T. Don't forget to specify which reward structure you want in the property.
 2. The performance of the system is actually a trade-off between several factors. In the sequel we study the power consumption of the system, assuming the energy is used at rate 0.13, 0.95 or 2.15 for SP power states "sleep", "idle" and "busy", respectively. In addition, transitions between power states have a fixed energy cost: going from "sleep" to "idle" uses 7.0 units and from "idle" to "sleep" uses 0.067 units. Add a reward structure to the model representing the power consumption. Use a cumulative reward property to investigate the energy consumption over time of the system.